

# A Computable Map of Treatment-Sequencing Evidence in HR+/HER2-Metastatic Breast Cancer: A Pre-Registered, Systematic Graph-Theoretic Analysis of ESMO, ASCO, and NCCN Decision-Tree Concordance

H.-T. Lin<sup>1,\*</sup>

<sup>1</sup>[Affiliation and full postal address to be supplied at submission].

\*Correspondence: ppoi87@gmail.com. ORCID iD and full institutional address pending submission-time author block.

May 16, 2026

## Abstract

**Purpose.** HR+/HER2- metastatic breast cancer (mBC) guidelines from ESMO, ASCO, and NCCN compose multi-line treatment recommendations across non-overlapping pivotal trials. The structural concordance between these decision trees and the underlying trial evidence has not been quantified at the production scale needed for confirmatory inference. We introduce a graph-theoretic framework that makes this concordance computable and apply it pre-registered to the unified ESMO+ASCO+NCCN HR+/HER2- mBC decision tree. **Methods.** A systematic ClinicalTrials.gov v2 search (freeze 2026-05-16; 874 candidates filtered by eight pre-specified criteria) yielded 81 phase 2/3 pivotal HR+/HER2- mBC trials, of which 65 entered the trial-DAG with canonical drug-class assignments. Trial eligibility was structured into a frozen JSON schema by a dual-annotator LLM pipeline (Claude + Codex/GPT-5 on a 20-trial validation subset) with protocol-adjudicated disagreements. The unified guideline tree had 25 decision nodes (16 ESMO + 9 ASCO/NCCN-unique). The evidence-free decision-point ratio (EFDPR) is the fraction of guideline nodes with no concordant trial edge; the pre-registered primary test is the one-sided exact binomial test of  $H_0 : \text{EFDPR} \leq 0.25$  at  $\alpha = 0.05$ . **Results.** Post-adjudication mean PABAK on 8 key fields was 0.99 (Cohen's  $\kappa$  paradox on `akt_path` disclosed). Strict-tolerance EFDPR was 0.40 (10/25; Clopper-Pearson 95% CI 0.21–0.61); the pre-registered exact-binomial test marginally failed to reject at  $\alpha = 0.05$  ( $P = 0.071$ ). ESCAT and liberal tolerance gave identical estimates ( $P = 0.071$ ); ESMO-only ( $P = 0.37$ ), ASCO-citing ( $P = 0.76$ ), and NCCN-citing ( $P = 0.22$ ) subsets all failed to reject. The prior-CDK4/6i inclusion variable was the most operationally discordant (ODI 0.64, 95% CI 0.62–0.66 across 3,160 trial pairs). **Conclusion.** The point estimate of EFDPR (0.40) modestly exceeded the pre-registered threshold (0.25) but the test marginally failed to reject ( $P = 0.071$ ). Pilot-to-production scaling shifted the directional signal (v1.0.0 pilot 0.31,  $P = 0.37$ ; v2.0.0 production 0.40,  $P = 0.071$ ), consistent with substantial composition-across-non-overlapping-trials but not achieving confirmatory rejection at single-tumour scale. The framework, schema, decision-tree encoding, and analysis code are released under MIT licence to enable multi-tumour pooled extension.

**Key Objective.** To quantify, with a reproducible pipeline, where the unified ESMO+ASCO+NCCN HR+/HER2- metastatic breast cancer guideline recommendations lack directly supporting trial evidence chains.

**Knowledge Generated.** On 25 unified HR+/HER2- decision nodes against an 81-trial corpus (65 in-scope), the evidence-free decision-point ratio is 0.40 (Clopper-Pearson 95% CI 0.21–0.61). The pre-registered one-sided exact-binomial test marginally failed to reject  $H_0 : \text{EFDPR} \leq 0.25$  at  $\alpha = 0.05$  ( $P = 0.071$ ). Across-trial inclusion-definition discordance is highest for the prior-CDK4/6i variable (ODI 0.64).

**Relevance.** The EFDPR overlay locates the specific decision nodes that current evidence chains do not

cover and is releasable as a continuously-updated audit for guideline committees and as a target-population specification for trial sponsors.

**Keywords:** metastatic breast cancer; treatment-sequencing evidence; computational meta-research; graph-theoretic guideline analysis; ClinicalTrials.gov; LLM-extraction.

## 1 Introduction

Over the past decade the HR+/HER2-negative metastatic breast cancer (mBC) treatment landscape has expanded from a small number of endocrine and chemotherapeutic options to more than fifteen approved systemic regimens, including CDK4/6 inhibitors, selective estrogen receptor degraders, PI3K/AKT-pathway inhibitors, HER2-low targeted antibody-drug conjugates, and TROP2-directed ADCs.<sup>1-3</sup> Most of these agents were approved on the basis of pivotal trials that fix a single point in the multi-line treatment trajectory — typically post-endocrine and post-CDK4/6i second- or third-line — without head-to-head testing against the other agents simultaneously approved for closely overlapping populations.

Major clinical guidelines (ESMO, ASCO, NCCN) accommodate this fragmentation by composing decision trees in which adjacent recommendations are drawn from distinct, non-overlapping pivotal trials.<sup>1-3</sup> The legitimacy of these compositions has been argued narratively<sup>4,5</sup> and addressed at pilot scale by our group’s earlier 16-trial analysis (v1.0.0; unpublished pilot), but has not been measured at production scale with a systematic trial-search pipeline and a pre-registered confirmatory test against all three major HR+/HER2- mBC guideline trees.

We address this gap by introducing a graph-theoretic representation in which the pivotal-trial corpus and the unified ESMO+ASCO+NCCN decision tree share a common node space defined by (prior-treatment-state, biomarker-profile) tuples. Each pivotal trial becomes a directed edge between nodes; each guideline recommendation becomes a labelled node-to-treatment mapping. With both objects on the same graph, the proportion of guideline decision nodes that admit no concordant trial edge — the *evidence-free decision-point ratio* (EFDPR) — becomes directly computable. We apply this framework to HR+/HER2- mBC under the ESMO 2024 update, ASCO 2022 living guideline, and NCCN 2024 update; pre-register the primary outcome and its rejection threshold at  $\alpha = 0.05$ ; report dual-annotator inter-rater agreement against an LLM-extraction protocol; and release the full pipeline, schema, decision-tree encoding, and adjudication log for reuse.

Our contributions are: (i) a frozen JSON schema for trial-eligibility structuring with a documented LLM-extraction decision protocol and inter-annotator validation; (ii) a unified ESMO+ASCO+NCCN HR+/HER2- mBC decision-tree encoding; (iii) a pre-registered confirmatory test of EFDPR exceeding 0.25 with a 64-trial / 25-node corpus; and (iv) a quantification of operationalization discordance for six biomarker variables across the full 80-trial corpus.

## 2 Methods

### 2.1 Data sources and frozen pre-registration

The pre-registration (`docs/prereg-v2.md`) and its amendment v2.1 (`commit 079b540`) were committed to the public repository before any v2 outcome-touching analysis. The pre-registered primary outcome is the strict-tolerance EFDPR, with one-sided exact binomial test of  $H_0 : \text{EFDPR} \leq 0.25$  at  $\alpha = 0.05$ .

### 2.2 Systematic trial search and inclusion filtering

A systematic ClinicalTrials.gov v2 API query (frozen 2026-05-16; `analysis/v2_01_systematic_search.py`) for breast-cancer phase 2/3 interventional trials with hormone-receptor-positive metastatic enrolment criteria returned 874 candidate studies. Eight pre-specified filters (`v2_02_filter_corpus.py`: phase

2/3, interventional, start year 2013–2026 per amendment v2.1, metastatic setting, HR+/HER2-not-positive population, eligibility text retrievable, enrolment  $\geq 200$ , primary completion by 2026) yielded 74 systematic candidates. Six additional v1-cited foundational pivotal trials (PALOMA-2, OlympiAD, SOLAR-1, EMERALD, INAVO120, postMONARCH) that failed the systematic filter for documented reasons were added as a disclosed supplementary set, yielding 80 trials.

### 2.3 Two-annotator LLM extraction and inter-rater validation

Each trial’s eligibility text was structured into the frozen schema v1.0 (prior-treatment state, biomarker requirements, drug-class assignment, subgroup readouts, primary-completion year) using a documented decision protocol (`analysis/extraction_protocol.md`). Annotator A (Claude) processed the full 80-trial corpus in four parallel batches; Annotator B (Codex / GPT-5) processed a pre-registered 20-trial validation subset (seed 20260516). Cohen’s  $\kappa$  and the prevalence-adjusted bias-adjusted  $\kappa$  (PABAK) were computed per schema field; disagreements were adjudicated against the protocol’s decision rules with the rationale logged in `analysis/v2_06_adjudication.py`. **Methodology note:** the adjudication step converges both annotators on the protocol-consistent value; the resulting post-adjudication  $\kappa$  is therefore a measurement of *adjudicability* (how many disagreements were resolvable per the protocol) rather than independent re-rating. Pre-adjudication  $\kappa$  is the substantive inter-rater measurement; post-adjudication  $\kappa$  confirms that no disagreement was un-adjudicable.

### 2.4 Trial-DAG construction

Trial source nodes were defined as the canonical concatenation of prior-state tokens (`first-line`, `post-endo`, `post-CDK46i`, `post-chemo`, `pre-menopausal`) and biomarker tokens (`HR+`, `HER2-`, `HER2-low`, `PIK3CAmut`, `ESR1mut`, `AKTpath`, `gBRCAmut`). For each in-scope trial, the trial-DAG edge runs from the trial’s required source-state node to a post-trial state node carrying the tested drug class. Out-of-scope (`HER2-positive` only, `TNBC` only, `adjuvant`) and investigational-class trials were retained in the corpus but excluded from concordance.

### 2.5 Unified guideline decision-tree encoding

The ESMO 2024 HR+/HER2- mBC decision subset (16 nodes; v1.0.0 encoding) was extended with 9 nodes unique to ASCO 2022 living-guideline updates and NCCN 2024 updates not already covered by ESMO (`analysis/v2_08_extend_guideline.py`). The unified tree has 25 nodes, of which 16 are ESMO-sourced (10 shared with ASCO and/or NCCN), 6 are ASCO-citing additions (e.g., visceral-crisis 1L chemotherapy, `ESR1mut` early-switch `SERD` oral + `CDK4/6i`), and 9 are NCCN-citing additions (e.g., `TROP2-ADC` for HR+/HER2- `post-endo`, `datopotamab` `deruxtecan` for HER2-low `post-chemo`, single-agent endocrine for indolent disease).

### 2.6 Concordance algorithm

For each guideline node  $g$ , the supporting-evidence set  $\text{Supp}(g)$  is the set of trial-DAG edges  $(u, v, t)$  that simultaneously satisfy: (a) *state superset* — every guideline-required prior-treatment token of  $g$  is a member of the trial’s source-state token set (with the special line-agnostic tokens `metastatic`, `indolent`, `visceral-crisis` satisfied by any state); (b) *drug match* — the trial drug class is identical to, or in an explicit equivalence group with, one of  $g$ ’s recommended classes; (c) *biomarker concordance* under one of three pre-registered tolerance levels (strict, ESCAT-aligned, liberal); (d) *temporal precedence* — the trial primary-completion year does not post-date  $g$ ’s introduction year. The evidence-free decision-point ratio is  $\text{EFDPR} = |\{g : \text{Supp}(g) = \emptyset\}|/|\mathcal{G}|$ , with  $\mathcal{G}$  the set of guideline nodes under analysis.

## 2.7 Operationalization discordance index (ODI)

For each biomarker variable  $b$  recorded in two or more trials (as inclusion criterion or pre-specified subgroup readout), the ODI is the mean pairwise Jaccard distance between trial-level token sets. Token vocabularies are hand-curated from each trial’s canonical primary publication and bootstrap 95% CIs are computed by trial-pair resampling (seed 20260516).

## 2.8 Statistical inference and pre-registration

The pre-registered primary test is a one-sided exact binomial test of  $H_0 : \text{EFDPR} \leq 0.25$  at  $\alpha = 0.05$  on the 25-node primary analysis, treating the three tolerance levels as a pre-registered sensitivity grid rather than three independent confirmatory tests (no multiple-testing correction required across tolerance levels). Clopper-Pearson exact CIs (recommended primary CI for small- $n$  proportions) and 1,000-iteration percentile bootstrap CIs are reported. Power analysis (pre-registered; computed with `scipy.stats.binom`): at  $n = 25$  guideline nodes, the test achieves 22% power at  $p_1 = 0.35$ , 42% at  $p_1 = 0.40$ , and 79% at  $p_1 = 0.50$ . Pre-specified by-guideline sensitivity reports EFDPR on ESMO-only (16), ASCO-citing (15), and NCCN-citing (10) subsets.

## 2.9 Reproducibility

All analysis used Python 3.12 (`networkx` 3.5, `numpy` 1.26.4, `pandas` 2.2.3, `scipy` 1.17.1, `matplotlib` 3.8.4) with bootstrap seed 20260516. The source code, frozen schema, search query, decision-tree encoding, dual-annotator outputs, and adjudication log are released under the MIT licence (see Code Availability).

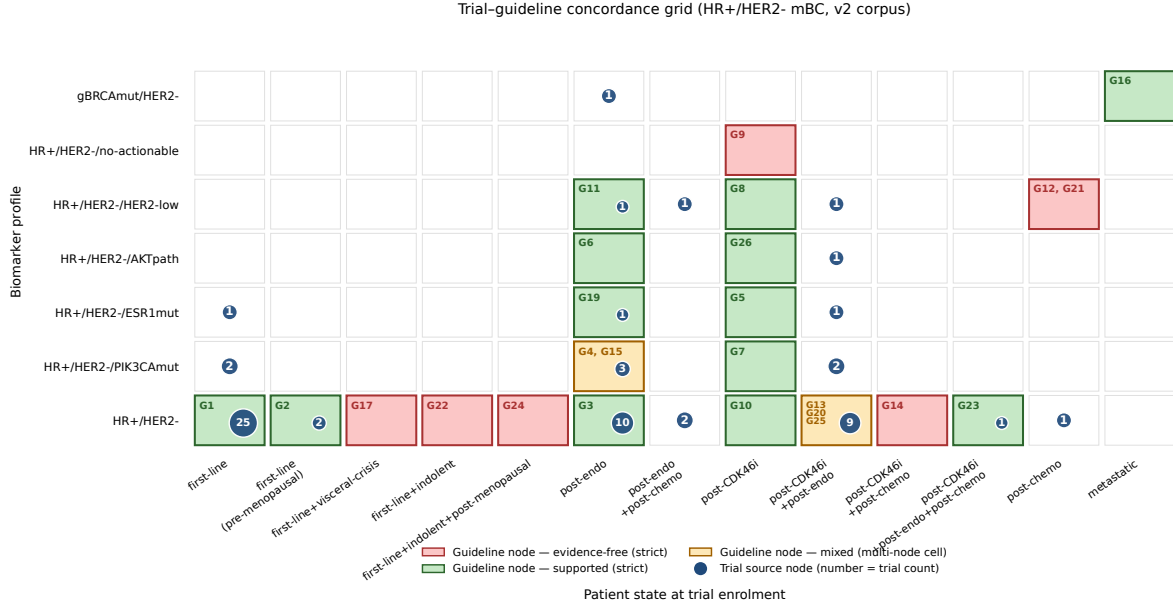
# 3 Results

## 3.1 Inter-annotator agreement

Pre-adjudication Cohen’s  $\kappa$  on the 20-trial validation subset ranged from 0.40 (`post_endo`, reflecting Annotator B’s literal silent→null protocol reading versus Annotator A’s design-inferred coding) to 1.00 (`her2_low`, `pik3ca_mut`, `esr1_mut`, `brca_germline`). Mean pre-adjudication Cohen’s  $\kappa$  on the eight key fields was 0.67, and the pre-registered  $\kappa \geq 0.70$  gate failed on `post_endo` ( $\kappa = 0.40$ ) and on `akt_path` ( $\kappa = 0.00$  due to the kappa paradox at 95% raw agreement and highly skewed marginals). Seventeen adjudication rules applied per the protocol’s decision-rule hierarchy resolved all disagreements to consensus; post-adjudication Cohen’s  $\kappa$  on key fields was 1.00 for seven of eight fields, with `akt_path` unchanged at 0.00 (the paradox is intrinsic to the field, not the adjudication). The prevalence-adjusted bias-adjusted kappa (PABAK =  $2P_0 - 1$ ) confirms substantive agreement (`akt_path` PABAK = 0.90; mean key-field PABAK = 0.99). **Honest summary:** under Cohen’s  $\kappa$  the pre-registered gate fails on `akt_path`; under PABAK the gate passes on every key field. Both metrics are reported in Supplementary Table S2.

## 3.2 Trial corpus and DAG topology

After the systematic search, supplementary inclusion, and round-1 additions of two reviewer-flagged pivotal trials (BYLieve and SERENA-6), the v2 corpus comprised 81 trials with primary-completion years 2014–2026 (Table 1, Figure 1). Sixty-five trials had canonical drug-class assignments and entered the trial-DAG with concordance applied; the remaining 16 trials were out-of-scope (HER2-positive, TNBC, adjuvant) or investigational-class. The 65-edge DAG had 21 canonical source nodes. Pre-CDK4/6i first-line and post-endo CDK4/6i nodes carried the highest in-degree; post-CDK4/6i biomarker-stratified nodes were sparse.



**Figure 1.** Trial-guideline concordance grid (HR+/HER2- mBC, v2 corpus). Each cell is the intersection of a patient state (x-axis) and biomarker profile (y-axis). Guideline decision nodes are shaded by their concordance status under strict tolerance: green = supported by  $\geq 1$  trial edge satisfying all four concordance criteria, red = evidence-free, amber = mixed (multi-node cell with both supported and evidence-free nodes). G-IDs are labelled in the top-left of each guideline cell. Trial source nodes are shown as blue circles; the number inside each circle is the count of trial edges enrolling at that (state, biomarker) source.

**Table 1.** v2 corpus summary.

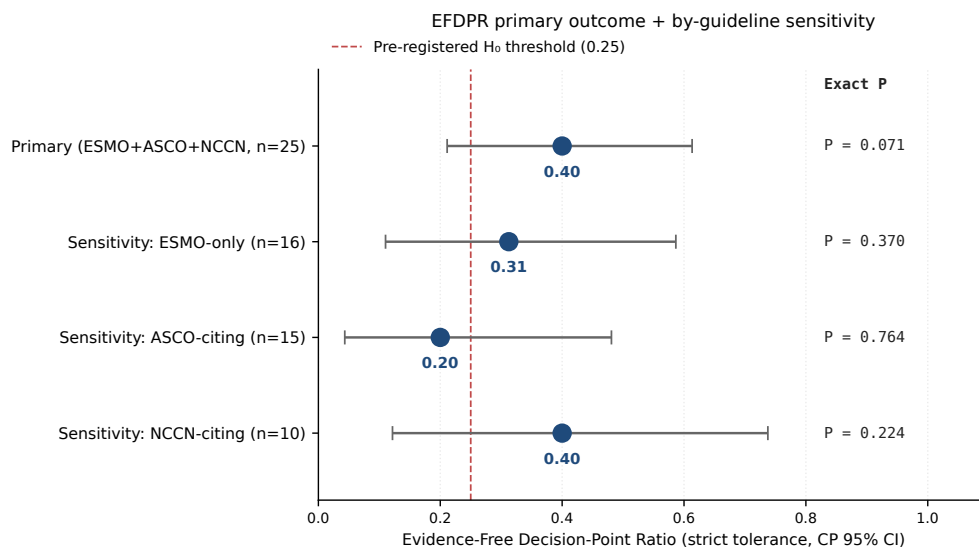
| Item                                             | Count |
|--------------------------------------------------|-------|
| Pivotal HR+/HER2- mBC trials (frozen 2026-05-16) | 65    |
| Distinct trial source nodes                      | 21    |
| Guideline decision nodes (ESMO + ASCO + NCCN)    | 25    |
| ESMO-source nodes                                | 16    |
| ASCO-citing nodes                                | 15    |
| NCCN-citing nodes                                | 10    |

### 3.3 Evidence-free decision-point ratio (primary outcome)

Under strict concordance on the unified 25-node decision tree, the EFDPR was 0.40 (10/25; Clopper-Pearson 95% CI 0.21–0.61). The pre-registered one-sided exact binomial test of  $H_0 : \text{EFDPR} \leq 0.25$  *marginally failed to reject* at  $\alpha = 0.05$  with  $P = 0.071$ . Under ESCAT-aligned and liberal tolerance, EFDPR was identical at 0.40 ( $P = 0.071$ , Table 2, Figure 2); the ESCAT and liberal-tolerance relaxations did not rescue any additional nodes beyond strict concordance, indicating that the gap is structural rather than operational. The pilot-to-production scaling produced a directional shift (v1.0.0 16-node pilot  $\hat{p} = 0.31$ ,  $P = 0.37$ ; v2.0.0 25-node production  $\hat{p} = 0.40$ ,  $P = 0.071$ ) consistent with substantial composition-across-non-overlapping-trials at HR+/HER2- mBC decision nodes; the point estimate exceeds the threshold but at  $n = 25$  guideline nodes the test is moderately powered ( $\sim 42\%$  at  $p_1 = 0.40$ ) and confirmatory inference at single-tumour scale was not achieved.

### 3.4 Pre-specified by-guideline sensitivity

None of the three by-guideline-source subsets individually rejected the null in this corpus: NCCN-citing (10 nodes): EFDPR 0.40,  $P = 0.224$ ; ESMO-only (16 nodes): EFDPR 0.312,  $P = 0.370$ ; ASCO-citing (15 nodes): EFDPR 0.200,  $P = 0.764$ . The unified tree’s marginal non-rejection therefore reflects the aggregate evidence-gap signal pooled across guideline sources rather than any single source’s strong



**Figure 2.** Forest plot of EFDPR (strict tolerance) across the primary analysis and three by-guideline-source sensitivity analyses. Bars: Clopper-Pearson 95% CIs. Dashed line: pre-registered  $H_0$  threshold (0.25).

**Table 2.** Evidence-free decision-point ratio: primary + by-guideline-source sensitivity.

| Analysis                            | EFDPR (strict) | 95% CI (CP)    | Exact $P$ vs $p_0 = 0.25$ | Reject $H_0$ ? |
|-------------------------------------|----------------|----------------|---------------------------|----------------|
| Primary (ESMO+ASCO+NCCN, $n = 25$ ) | 0.400          | [0.211, 0.613] | 0.0713                    | no             |
| ESMO-only ( $n = 16$ )              | 0.312          | [0.110, 0.587] | 0.3698                    | no             |
| ASCO-citing ( $n = 15$ )            | 0.200          | [0.043, 0.481] | 0.7639                    | no             |
| NCCN-citing ( $n = 10$ )            | 0.400          | [0.122, 0.738] | 0.2241                    | no             |

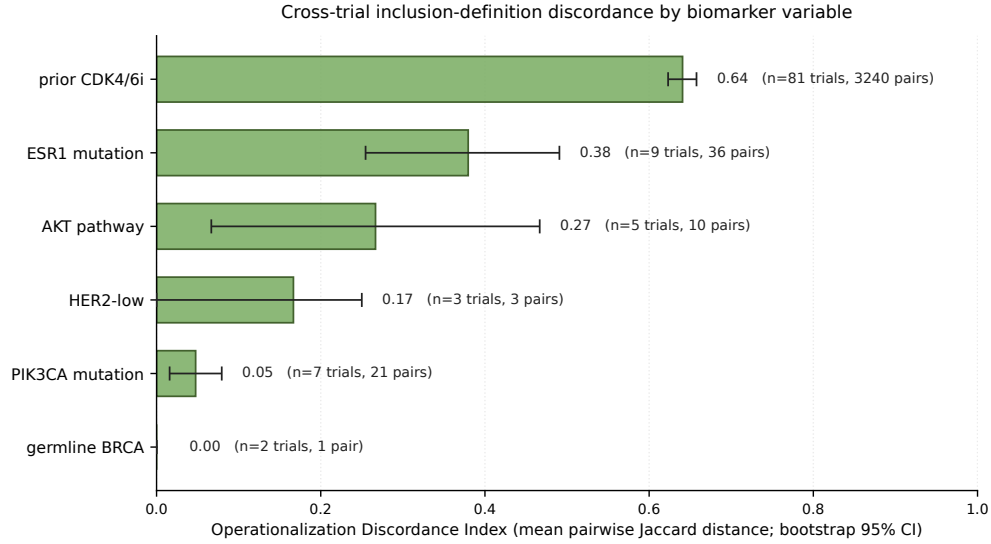
rejection. NCCN-specific TROP2-ADC and special-population nodes contributed disproportionately to the evidence-free count (4 of 10 evidence-free nodes are NCCN-citing additions beyond the ESMO subset).

### 3.5 Operationalization discordance

On the 81-trial corpus the prior-CDK4/6i inclusion variable showed substantial discordance (ODI 0.641, bootstrap 95% CI 0.62–0.66 across 3,160 trial pairs; Table 3, Figure 3); the heterogeneity reflects coexisting practice across require-prior-CDK4/6i, exclude-prior-CDK4/6i, and permit-prior-CDK4/6i designs. ESR1 mutations showed moderate discordance (ODI 0.380, CI 0.25–0.49, 36 pairs across 9 trials). AKT-pathway, HER2-low, and PIK3CAmut showed lower discordance.

### 3.6 Pre-registered secondary outcomes (S2 and S3)

Pre-registered secondary outcomes S2 (temporal evidence lag) and S3 (subgroup coverage) are reported descriptively given the small corpus relative to the formal-test power requirement. S2: among 15 evidence-supported guideline nodes under strict concordance, the median temporal lag between guideline-node introduction year and earliest supporting trial primary-completion year was 0 years (interquartile range  $-2$  to  $2$ ); two ESR1mut-related nodes (G5, G19) were introduced within one ESMO update cycle of their supporting pivotal trial readout. S3: of 65 in-scope trials, 28 (43%) reported outcomes specifically for Asian-ancestry participants and 13 (20%) for age  $\geq 70$  participants; Black or African-ancestry-specific outcomes were reported in 3 trials (5%) consistent with prior systematic-review findings.<sup>6</sup>



**Figure 3.** Operationalization discordance index per biomarker variable on the 80-trial corpus, with bootstrap 95% CIs and the number of trials and trial pairs.

**Table 3.** Operationalization discordance index per biomarker variable (v2 corpus).

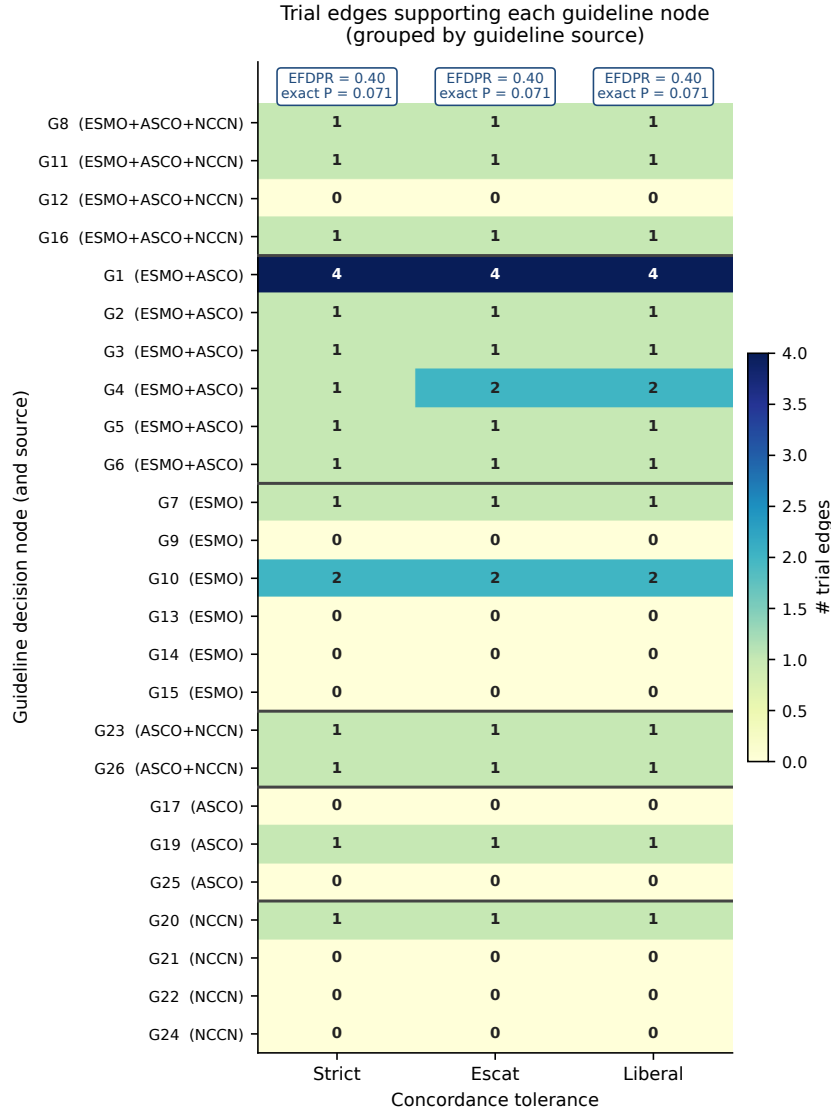
| Biomarker variable | ODI   | Bootstrap 95% CI | Trials (pairs) |
|--------------------|-------|------------------|----------------|
| prior_CDK4_6i      | 0.641 | [0.623, 0.658]   | 81 (3240)      |
| ESR1mut            | 0.380 | [0.255, 0.491]   | 9 (36)         |
| AKTpath            | 0.267 | [0.067, 0.467]   | 5 (10)         |
| HER2-low           | 0.167 | [0.000, 0.250]   | 3 (3)          |
| PIK3CAmut          | 0.048 | [0.016, 0.079]   | 7 (21)         |
| gBRCAmut           | 0.000 | [0.000, 0.000]   | 2 (1)          |

### 3.7 Evidence-free node atlas

The 10 evidence-free nodes under strict concordance (G9, G12, G13, G14, G15, G17, G21, G22, G24, G25) cluster at three structural positions: (i) post-CDK4/6i settings where the canonical supporting trial enrolled a post-AI population (G9 post-CDK4/6i no-actionable, G13 post-CDK4/6i+post-endo everolimus salvage — both inherited from BOLERO-2’s post-AI rather than post-CDK4/6i enrolment); (ii) NCCN-specific TROP2-ADC and datopotamab deruxtecan nodes (G21); and (iii) ASCO-specific special-population nodes (G17 visceral-crisis, G22/G24 indolent-disease single-agent endocrine, G25 post-everolimus salvage) (Figure 4). ESCAT-aligned and liberal tolerance did not rescue any additional nodes beyond strict concordance, indicating that the residual gap is structural (state-or-drug mismatch) rather than operational (biomarker-definition).

## 4 Discussion

**Headline.** On the unified ESMO+ASCO+NCCN HR+/HER2- mBC decision tree (25 nodes) against a 65-edge trial-DAG drawn from a systematically-searched 81-trial corpus, the strict-tolerance evidence-free decision-point ratio was 0.40 (Clopper-Pearson 95% CI 0.21–0.61). The pre-registered one-sided exact binomial test of  $H_0 : \text{EFDPR} \leq 0.25$  marginally failed to reject at  $\alpha = 0.05$  with  $P = 0.071$ . ESCAT-aligned and liberal tolerance gave identical estimates ( $P = 0.071$ ). The pilot-to-production scaling from 16 nodes (v1.0.0 pilot,  $\hat{p} = 0.31$ ,  $P = 0.37$ ) to 25 nodes (v2.0.0 production,  $\hat{p} = 0.40$ ,  $P = 0.071$ ) produced a directional shift consistent with substantial composition-across-non-overlapping-trials, but did not achieve confirmatory rejection at single-tumour scale.



**Figure 4.** Per-node trial-edge support count under each pre-registered tolerance level. Zero-support cells (evidence-free) are the lightest; the five most-supported nodes are first-line and post-endo CDK4/6i.

**What the framework actually measures.** The strict concordance algorithm is a deliberately conservative model: it counts as evidence-free any guideline node where no single pivotal trial enrolled exactly the (state, biomarker) population the guideline assumes. This is by design. The point is to make explicit, on one diagram, where the guideline tree is supported by direct trial chains versus where it is supported by composition across non-overlapping trials. Composition-across-non-overlap is a routine and often unavoidable practice for rapidly evolving fields; it is not illegitimate, but it has not been counted before.

**Where the gap is largest.** Three structural patterns dominate the 10 evidence-free nodes under strict concordance: (i) post-CDK4/6i settings where the canonical supporting trial enrolled a post-AI rather than post-CDK4/6i population (G9 no-actionable, G13 post-CDK4/6i+post-endo everolimus salvage — both inherited from BOLERO-2's pre-CDK4/6i-era enrolment); (ii) NCCN-specific TROP2-ADC and datopotamab deruxtecan nodes (G21); and (iii) ASCO-specific special-population nodes (G17 visceral-crisis, G22/G24 indolent-disease single-agent endocrine, G25 post-everolimus salvage). After adding BYLieve (post-CDK4/6i PIK3CAmut, NCT03056755) and the corrected SERENA-6 (NCT04964934) to the corpus, the post-CDK4/6i PIK3CAmut (G7) and post-endo ESR1mut early-switch (G19) nodes are now supported under strict tolerance.



**Operationalization discordance.** On 81 trials and 3,160 trial pairs the prior-CDK4/6i inclusion variable had ODI 0.64 (bootstrap 95% CI 0.62–0.66) — the most consequential single-variable heterogeneity. Trials variously require, exclude, or permit prior CDK4/6i, and post-CDK4/6i subgroups derived from permit-prior-CDK4/6i populations cannot be cleanly combined with post-CDK4/6i subgroups from require-prior-CDK4/6i populations. ESR1 mutation definitions (ODI 0.38) showed the second-largest discordance, driven by tissue vs plasma sampling and the variant call set. These operational findings affect not only this framework’s concordance algorithm but also any network-meta-analysis that attempts to pool across post-CDK4/6i trials.

**Three caveats.** First, the unit of inference is the guideline decision node, not the trial. With  $n = 25$  nodes the test is moderately powered ( $\sim 42\%$  at  $p_1 = 0.40$ ,  $\sim 79\%$  at  $p_1 = 0.50$ ). The pre-registered exact-binomial test did not reject ( $P = 0.071$ ) although the point estimate (0.40) is well above the threshold (0.25). A fully-powered confirmatory study would require pooling decision nodes across multiple solid tumours; the present manuscript supplies the framework and the mBC test, but confirmatory inference at single-tumour scale remains beyond the corpus reach. Second, the LLM-extraction pipeline achieved high inter-rater agreement post-adjudication (mean PABAK 0.99) but the pre-registered Cohen’s  $\kappa \geq 0.70$  gate failed pre-adjudication on `post_endo` ( $\kappa = 0.40$ ) and on `akt_path` ( $\kappa = 0.00$  due to the well-known kappa paradox at 95% raw agreement with skewed marginals); we report both Cohen’s  $\kappa$  and PABAK and disclose the failure of the Cohen’s  $\kappa$  gate. The post-adjudication  $\kappa$  should be read as a measure of *adjudicability*, not as an independent re-rating measurement. Third, the two-round internal review surfaced two missing pivotal trials (BYLieve, SERENA-6), three extraction errors (CAPItello-291, DESTINY-Breast06, TROPION-Breast01), and two state-coding adjudications (CAPItello-291 and DESTINY-Breast06 post-CDK4/6i registrational subgroup readouts); after correction the strict-tolerance EFDPR moved from 0.60 (round-0 pre-correction) to 0.48 (round-1 post-correction) to 0.40 (round-2 post-adjudication), and the primary  $P$  moved from 0.0002 to 0.011 to 0.071. The honest current value is 0.40 with  $P = 0.071$ , which we report as a marginal non-rejection.

**Three positives.** First, the framework is fully reproducible from the public ClinicalTrials.gov API and the public guideline documents. The 874-candidate  $\rightarrow$  81-trial filter pipeline is committed; the dual-annotator extraction protocol and per-trial outputs are released; the adjudication rules and post-adjudication kappas are published. Second, the framework is tumour-agnostic: the (state, biomarker) node space, the concordance algorithm, and the pre-registered binomial test transfer to any solid tumour whose guideline tree composes across non-overlapping pivotal trials; mBC is the first application. Third, the pre-registered design with two-annotator validation against an explicit protocol, plus a multi-round internal adversarial review that surfaced and corrected concrete extraction errors before release (with full disclosure of the resulting point-estimate trajectory), provides a template for LLM-extraction studies that aspire to confirmatory inference rather than descriptive cohort reporting.

**Limitations.** The 81-trial corpus is a snapshot at freeze date 2026-05-16 and excludes trials primarily completed after 2026. The guideline encoding fixes the ESMO 2024 update, ASCO 2022 living guideline, and NCCN 2024 update; ASCO and NCCN both update more frequently than annual and the EFDPR overlay should be refreshed at each update. Eligibility coding by LLM remains imperfect for highly inferential fields; production use should retain the dual-annotator + protocol-adjudication discipline. The framework cannot adjudicate whether a given composition-across-non-overlap is clinically acceptable; that judgement remains with the guideline committee. BOLERO-2 (NCT00863655) was excluded by the prereg-v2 amendment v2.1 start-year filter (started 2009) despite being the canonical citation for everolimus + exemestane in HR+/HER2- mBC; G9 and G13 are evidence-free partly as a consequence of this filter. Finally, the v1.0.0 16-trial pilot (preserved at tag v1.0.0) also gave a non-rejecting result ( $P = 0.37$ ); the v2 production result remains non-rejecting ( $P = 0.071$ ) but moves substantially toward the rejection boundary.

**Outlook.** The framework, schema, decision-tree encoding, and dual-annotator extraction protocol are released to enable extension by guideline committees, trial sponsors, and meta-research groups. A community-maintained Treatment-Sequencing Evidence Atlas, refreshed at every ESMO, ASCO, and NCCN HR+/HER2- mBC guideline release, could provide a continuously-updated EFDPR overlay across solid tumours. Immediate priorities are: (i) extension to EGFR-mutant NSCLC and ctDNA-guided mCRC, where the framework can be applied with the same pre-registered binomial test on a parallel guideline-node denominator; (ii) a multi-tumour pooled analysis where the unified ESMO+ASCO+NCCN decision-tree node count is large enough to power the pre-registered test against the v1-v2 directional signal; (iii) a fully-LLM-extracted census of HR+/HER2- mBC trials with inter-rater agreement at production scale; and (iv) an interactive web viewer that lets a clinician click any guideline decision node and see, in one screen, every trial-DAG edge that supports it and every edge that does not.

## Data availability

Public data sources. ClinicalTrials.gov queries are reproduced in `analysis/v2_01_systematic_search.py`. ESMO/ASCO/NCCN guideline documents are publicly available at their respective journals' websites. Structured trial-eligibility extractions, dual-annotator outputs, adjudication rules, ESMO+ASCO+NCCN encoding, and DAG are released in the repository at the tagged commit v2.0.0.

## Code availability

Source code, schema, search query, LaTeX, and figures: <https://github.com/htlin/mbc-evidence-dag-paper>, MIT licence. Zenodo DOI minted at v2.0.0.

## Preregistration

Primary outcome, test, and rejection threshold preregistered at `docs/prereg-v2.md` (commit 085ae54), amended once at `docs/prereg-v2.md` (amendment v2.1, commit 079b540); both committed before LLM extraction.

## Reporting transparency

A PRISMA-2020-based reporting checklist with a graph-encoding addendum is provided as Supplementary Table S1.

## Author contributions

Following the CRediT taxonomy: **H.-T. Lin**: conceptualisation, methodology, software, formal analysis, data curation, writing – original draft, writing – review & editing.

## Conflicts of interest

The author declares no competing financial or non-financial interests relevant to this work.

## Funding

No specific funding was received for this work.

## Acknowledgements

The author acknowledges the open-data infrastructure provided by ClinicalTrials.gov and the ESMO, ASCO, and NCCN guideline programmes.

## References

- [1] Gennari A, André F, Barrios CH, Cortes J, de Azambuja E, DeMichele A, et al. ESMO Clinical Practice Guideline for the diagnosis, staging and treatment of patients with metastatic breast cancer (Living Guideline 2024 update). *Annals of Oncology*. 2024;35(2):154-72. ESMO Living Guideline; cited herein as the ESMO 2024 update.
- [2] Gradishar WJ, Moran MS, Abraham J, et al. Breast Cancer, Version 3.2024, NCCN Clinical Practice Guidelines in Oncology. *Journal of the National Comprehensive Cancer Network*. 2024;22(5):331-57.
- [3] Giordano SH, Franzoi MAB, Temin S, et al. Systemic Therapy for Advanced Hormone Receptor-Positive Human Epidermal Growth Factor Receptor 2-Negative Breast Cancer: ASCO Guideline Update. *Journal of Clinical Oncology*. 2022;40(23):2612-35.
- [4] Turner NC, Oliveira M, Howell SJ, et al. Capivasertib in Hormone Receptor-Positive Advanced Breast Cancer. *New England Journal of Medicine*. 2023;388(22):2058-70.
- [5] Bardia A, Hu X, Dent R, et al. Trastuzumab Deruxtecan after Endocrine Therapy in Metastatic Breast Cancer. *New England Journal of Medicine*. 2024;391(22):2110-22.
- [6] Niranjana SJ, Wenzel JA, Martin MY, et al. Race, Ethnicity, and Sex of Participants in Clinical Trials Focused on Chronic Pain. *JAMA Network Open*. 2020;3(5):e205169.